

A Third-Order Memory Hénon Map: An Exact Low-Period Pilot

Anonymous Route-A report

Abstract

We study the three-dimensional recurrence $G(x, y, z) = (x^2 - 55/16 - y - z/2, x, y)$, in which two delay coordinates are part of the state. Exact symbolic elimination gives two fixed points and one genuine primitive period-two cycle. The ordered Jacobian product has determinant $1/4$ and trace $-15/4$. Degree growth $2, 4, 8$ shows why these finite monodromy data do not automatically define a polynomial Fredholm owner.

1 Map and fixed locus

The map is

$$G(x, y, z) = (x^2 + a - y - \kappa z, x, y), \quad (a, \kappa) = (-55/16, 1/2).$$

Its Jacobian at a state with first coordinate u is

$$J(u) = \begin{pmatrix} 2u & -1 & -1/2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \det J(u) = -1/2.$$

The fixed equations force $x = y = z$ and give

$$x = 5/4 - \sqrt{5}, \quad x = 5/4 + \sqrt{5}.$$

2 Primitive two-cycle and monodromy

Direct substitution gives the exact cycle

$$p_0 = (-7/4, 1/4, -7/4) \mapsto p_1 = (1/4, -7/4, 1/4) \mapsto p_0.$$

It is not fixed. With $M_2 = J(1/4)J(-7/4)$, exact multiplication gives

$$M_2 = \begin{pmatrix} -11/4 & -1 & -1/4 \\ -7/2 & -1 & -1/2 \\ 1 & 0 & 0 \end{pmatrix}, \quad \det M_2 = 1/4, \quad \text{tr } M_2 = -15/4,$$

and

$$\det(\lambda I - M_2) = \frac{4\lambda^3 + 15\lambda^2 - 2\lambda - 1}{4}.$$

The fixed-point characteristic polynomials are recorded in the exact evidence receipt and independently recomputed by SymPy.

Proposition 1. *The fixed locus, primitive two-cycle, Jacobian determinants, and displayed monodromy polynomial are reproduced by an independent symbolic checker.*

Proof. The checker solves the three fixed equations, evaluates both directions of the proposed cycle, and recomputes the ordered Jacobian product without importing producer functions. $\square$

3 Memory and Route-A boundary

The first coordinate degrees after three forward iterates are $2, 4, 8$. Thus a finite polynomial-degree truncation is not invariant by inspection; any analytic transfer construction would need a named function space and a tail estimate. The finite ledger earns ‘A1_WEAK’ and ‘A2_CERTIFIED_PREFIX’, with ‘A3_NOT_ADDRESSED’ and ‘A4_FAIL’. No complete three-dimensional orbit atlas, arithmetic data, analytic Fredholm determinant, or Route-B object is claimed. The scope firewall is ‘NO_BAD_EULER_OR_ROOT_NUMBER’.